

Multimodal Deep Learning Framework Integrating Satellite Imagery and Reanalysis Climate Data for Predicting Regional Heat Wave and Flash Flood Occurrences

Project Documentation Overview

This document summarizes the implementation completed for the project. It is intended as a concise overview suitable for project discussions and viva preparation.

1. Project Objective

Develop a multimodal deep learning framework that combines satellite imagery with ERA5 reanalysis weather data and MODIS land surface temperature to classify Heatwave and Flash Flood events.

2. Workflow

1. Collect historical events.
2. Download Sentinel-2/Landsat images.
3. Extract CNN features using ResNet18.
4. Extract ERA5 weather variables.
5. Extract MODIS LST.
6. Merge datasets and preprocess.
7. Train Transformer on weather features.
8. Fuse CNN and Transformer features using Cross-Attention.
9. Evaluate the model.
10. Perform inference.

3. Important Notebooks

- 01 Event Filtering
- 02 Image Download
- 03 Feature Extraction
- 05A CNN DataLoader
- 05B ResNet18 Training
- 05C CNN Feature Extraction
- 06A ERA5 Extraction
- 06B MODIS Extraction
- 07 Data Preprocessing
- 08 Transformer Encoder
- 09 Cross Attention Fusion
- 10 Model Evaluation
- 11 Model Inference

4. Generated CSV Files

- image_metadata.csv
- cnn_features.csv
- era5_features.csv
- modis_features.csv
- training_dataset.csv
- transformer_features.csv

5. Saved Models

cnn_encoder_resnet18.pth
weather_transformer.pth
multimodal_model.pth

6. Architecture

Satellite Images -> ResNet18 -> CNN Features
ERA5 + MODIS -> Transformer -> Weather Features
CNN Features + Weather Features -> Cross Attention -> Classification

7. Evaluation

The trained model achieved perfect performance on the available held-out test split (100% accuracy, precision, recall and F1). Because the dataset contains only 32 matched events, these results should be presented as proof of implementation rather than evidence of real-world generalization.

8. Future Scope

Replace historical ERA5 inputs with live weather forecasts (e.g. GFS/Open-Meteo), automate satellite image retrieval, retrain the weather branch using forecast-compatible variables, and deploy as a web or mobile application.